Face Image Quality Assessment (FIQA) System

Engineering Status, Architectural Deliverables & Technical Report

Project: eDifFIQA & UniFace Pipeline	**Author / Engineer:** Deepak
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Status: Core Architecture & Localhost Complete	**Deliverable:** Video Walkthrough + Codebase

Executive Summary

This project implements a complete, end-to-end **Face Image Quality Assessment (FIQA)** application designed to evaluate face biometrics, predict utility scores, and enforce rigorous quality gates for biometric enrollment and surveillance verification. The system combines deep neural networks (eDifFIQA-S, SCRFD, RetinaFace, 6DRepNet, MobileGaze) with explicit photographic diagnostics and a reactive, user-friendly React/Vite dashboard backed by FastAPI.

Current Status: The complete application is **100% operational on the local environment**, with real-time multi-face detection, quality scoring, landmark alignment, pose/gaze estimation, and batch ZIP/CSV exports verified. A full demo video has been recorded showcasing end-to-end execution. Cloud deployment to public URLs has been implemented, with technical optimizations applied to address free-tier memory constraints.

1. Modules & Capabilities Delivered (What is Done)

Component	Technical Specification	Status
eDifFIQA-S Scoring	Pretrained ResNet backbone predicting ISO/IEC compliant quality utility scores (0.0 to 1.0).	**COMPLETED**
Multi-Stage Detection	UniFace SCRFD as primary detector + RetinaFace fallback with IoU deduplication for crowded scenes.	**COMPLETED**
Facial Alignment	5-point facial landmark affine transformation generating normalized 112×112 face crops.	**COMPLETED**
Head-Pose Estimation	UniFace 6DRepNet estimating real 3D Euler angles (Pitch, Yaw, Roll) with configurable thresholds.	**COMPLETED**
Gaze Estimation	UniFace MobileGaze estimating pitch and yaw gaze vectors to detect direct camera focus.	**COMPLETED**
Explainable Diagnostics	Laplacian variance (blur/sharpness), exposure histograms (under/overexposure), and illumination asymmetry.	**COMPLETED**
6-State Gate Engine	PASS, REVIEW, REJECT, NO_FACE, NOT_EVALUATED, ERROR based on calibrated operational thresholds.	**COMPLETED**
FastAPI Backend	Asynchronous chunk streaming, dynamic threshold calibration API, session stats, and latency metrics.	**COMPLETED**
React/Vite Dashboard	Single-image inspection, landmark visualization, live batch progress bar, and 1-click ZIP/CSV export.	**COMPLETED**

2. Technical Challenges Faced & Engineering Solutions

Challenge A: Missed Faces in Dense / Crowded Surveillance Photos

• *Issue:* Standard single-stage detectors missed smaller, partially occluded, or boundary-adjacent faces in crowded tourist images.
• *Solution:* Implemented a two-stage hybrid detector chain. UniFace SCRFD acts as primary for speed, while RetinaFace runs as a complementary fallback. An IoU-based deduplication algorithm merges overlapping bounding boxes without suppressing adjacent distinct individuals.

Challenge B: Deep Learning Multi-Model RAM Footprint vs. 512 MB Cloud Limits

• *Issue:* Running 5 neural networks simultaneously (SCRFD + RetinaFace + eDifFIQA + 6DRepNet + MobileGaze) in PyTorch/ONNX on CPU naturally requires **~700–830 MB of RAM**. On free-tier cloud containers (such as Render) with a hard 512 MB ceiling, batch inference triggered the Linux Out-Of-Memory (OOM) killer, resulting in HTTP 502 Bad Gateway errors.
• *Solutions Applied:*
1. **Shared Singleton Weights:** Prevented redundant duplicate 200MB model re-loads in QualityPipeline.
2. **Kernel Memory Trimming:** Injected ctypes.CDLL('libc.so.6').malloc_trim(0) to immediately return unmapped memory to the OS.
3. **Micro-Chunk Streaming:** Configured the React batch client to stream files in micro-chunks of 2, bounding memory spikes.
4. **Safe Pre-Scaling:** Automatically downscaled 4K/oversized images to 1920px max dimension, cutting raw bitmap memory by >80%.
5. **Dual Deployment:** In addition to Render, deployed a public instance on **Hugging Face Spaces (16 GB Free RAM)** where large concurrent multi-face batches run without any memory constraints.

3. Current Status Matrix & Scope Overview

Environment / Capability	Observed Performance & Behavior	Current State
Localhost Environment	Full functionality, high speed, unlimited batch capacity, all 5 models run concurrently. Demo video captured.	**100% READY**
Hugging Face Spaces (16 GB RAM Tier)	Runs full multi-model pipeline without memory bottlenecks. Ideal public platform for stakeholder review.	**DEPLOYED & LIVE**
Render Cloud (512 MB Free Tier)	Operational for single images and lightweight batches. Heavy multi-model batches require high-RAM tier or light preset.	**CONSTRAINED**
Operational Threshold Calibration	Engine supports real-time threshold updates via API/UI. Final calibration pending customer domain dataset.	**IN PROGRESS**

4. Next Steps & Recommendations

1. **Video Review:** Review the recorded video demonstration showcasing localhost performance.
2. **Customer Dataset Calibration:** Establish domain-specific quality score acceptance/rejection thresholds based on Graymatics target camera feeds.
3. **Production Hosting Architecture:** For production deployment of multi-face batches, a compute instance with ≥2

GB RAM is recommended to support all 5 models comfortably under peak load.